



Adaptive gate residual connection and multi-scale RCNN for fake news detection

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ABSTRACT

Detection of false news based on text classification technology has significant research significance and practical value in the current information age. However, existing methods overlook the problem of uneven sample distribution in the false news dataset and fail to consider the mutual influence between news articles. In light of this, this paper proposes a new method for false news detection. Firstly, news texts are embedded using Electra (Efficiently Learning an Encoder that Classifies Token Replacements Accurately) to obtain word embedding representations. Secondly, Multi-Scale Recurrent Convolutional Neural Network (RCNN) is employed to further extract contextual information from news texts. Self-attention is introduced to calculate attention scores between news articles, allowing for mutual influence between news features. The establishment of connections between modules is achieved through adaptive gated residual connections. Finally, the focal loss function is used to balance the relationship between few-sample and multi-sample data in the dataset. Experimental results on publicly available false news detection datasets demonstrate that the proposed method achieves higher prediction accuracy than the comparative methods. This method provides a new perspective for the field of false news detection, playing a positive role in promoting information authenticity and protecting public interests.

1. Introduction

In today's information age, the speed and scope of news dissemination are unprecedented, allowing people to access various information anytime and anywhere through the internet (Capuano, Fenza, Loia, & Nota, 2023). However, with the rise of social media and online platforms, the spread of fake news has become increasingly serious (Hangloo & Arora, 2021). Fake news not only misleads the public and undermines the authenticity and reliability of information but also poses a threat to social stability and national security. Currently, how to effectively identify and filter fake news has become an urgent problem. Based on the above, this paper proposes a fake news detection method based on adaptive gated residual connections and focal loss, referred to as AGRC-RCNN (Adaptive Gate Residual Connection and Multi Scale RCNN), which aims to improve the accuracy of fake news detection tasks.

Fake news detection based on text classification (Kowsari et al., 2019) technology provides an effective solution. Text classification is a core technology in the field of natural language processing (NLP), which automatically categorizes text content using computer algorithms to identify the theme, sentiment, intent, etc. Applying text

classification technology to fake news detection can help us automatically identify and filter out false information, thereby improving the quality and credibility of information.

In text classification technology, the quality of word embeddings directly determines the final predictive performance, and good feature engineering is a prerequisite for achieving high performance (Minaee et al., 2021). In the early days, the most commonly used were static language models such as Word2vec (Word to Vectors) and GloVe (Global Vectors for Word Representation), which had the disadvantage of being unable to express polysemy (Wang et al., 2024). Later, large-scale dynamic language models such as BERT (Bidirectional Encoder Representations from Transformers) (Korotееv, 2021) based on self-attention mechanisms and pre-training patterns emerged. BERT has brought a new perspective to NLP, as excellent NLP performance can be achieved through pre-training in a general domain combined with downstream task fine-tuning. Therefore, most existing fake news detection methods are based on models like BERT as word embedding models.

Deep learning networks like RNNs (Recurrent Neural Networks) and CNNs (Convolutional Neural Network) are commonly used feature

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extraction tools in text classification (Zulqarnain, Ghazali, Hassim, & Rehan, 2020). Networks like LSTM (Long Short-Term Memory) and GRU (Gated Recurrent Unit), belonging to the RNN category, excel at capturing contextual information from text, while networks like TextCNN, belonging to the CNN category, are adept at extracting local feature information from text (Wang, He, Zhang, & Liu, 2020). Therefore, existing methods for detecting fake news often employ RNNs or CNNs to further extract feature information from text.

The self-attention mechanism (Wang, Chen, Liu, & Tai, 2022), also known as internal attention mechanism, is an important technique in deep learning, especially playing a crucial role in text classification tasks in the field of natural language processing. The core idea of self-attention mechanism is to enable the model to automatically learn the relationships between elements within a sequence data (such as text), thereby capturing richer contextual information (Fang et al., 2021). In order to focus the model's attention on important parts of news text during prediction, existing news detection methods often calculate self-attention on the news text before making predictions.

Based on the above analysis of existing work in fake news detection, we identify three key challenges that need to be addressed:

(1) Current methods largely rely on single-scale feature extraction, which may limit the model's ability to capture the rich contextual information at different granularities in news text. While deep learning networks like RNNs and CNNs have shown promise in text feature extraction, they typically operate at a fixed scale, potentially missing important multi-scale patterns.

(2) Most existing approaches process each news article independently, overlooking the valuable inter-article relationships. In real-world scenarios, the veracity of news is often determined by cross-referencing multiple related articles. The self-attention mechanism, which has proven effective in capturing internal text relationships, could be adapted to model these cross-article dependencies.

(3) The inherent class imbalance in fake news datasets poses a significant challenge, as real news typically outnumbers fake news. Additionally, the gradient vanishing problem in deep networks can impair the model's ability to learn effectively from both majority and minority classes.

To address these challenges, this paper proposes an integrated approach that combines multi-scale feature extraction, cross-article attention modeling, and adaptive residual learning. Specifically, this paper has the following highlights:

- This paper introduces a Multi-Scale Recurrent Convolutional Neural Network (MSRCNN) to extract contextual information from news texts at various angles and scales, thereby enriching the informational scope of the news texts.
- This paper proposes an adaptive gated residual connection module, which can autonomously determine how much residual information needs to be considered.
- The proposed method achieves state-of-the-art performance on three public fake news detection datasets.

2. Related work

2.1. Fake news detection based on machine learning

Khanam, Alwasel, Sirafi, and Rashid (2021) address fake news detection on social media using machine learning algorithms like Random Forest, Support Vector Machine, and Naive Bayes. These algorithms employ feature extraction, vectorization, and feature selection to analyze text data and achieve high detection accuracy. Baarir and Djeflal (2021) propose a machine learning-based fake news detection system. This system combines the bag-of-words model and n-gram techniques, and applies Term Frequency-Inverse Document Frequency (TF-IDF) (Zhou, 2022) for feature extraction. SVM is then used as the classifier to predict fake news. Probiez, Stefański, and Kozak (2021) propose a machine

learning approach for detecting fake news, using TF-IDF for text vectorization, followed by Random Forest and SVM models for predictions. This method enhances fake news detection, especially for large-scale social media data, enabling quick and efficient misinformation identification. Bharath, Manikanta, Prakash, Sumathi, and Chinnasamy (2021) propose a model for detecting fake news on Twitter by comparing the performance of five machine learning algorithms: SVM, Naive Bayes, Logistic Regression, and RNN. Experimental results indicate that SVM and Naive Bayes classifiers demonstrate superior classification accuracy compared to the other models. Hakak et al. (2021) introduce an ensemble model for fake news classification that combines Decision Tree, Random Forest, and Extra Tree Classifier (ET) to boost accuracy. Key features like word count, character count, sentence count, average word length, and sentence length are used to enrich the feature set and improve classification precision.

Machine learning-based fake news detection methods, using models like SVM, random forest, and decision trees, offer fast predictions with low resource consumption and perform well on small datasets (Gasparetto, Marcuzzo, Zangari, & Albarelli, 2022). However, they struggle with large datasets and complex tasks, often relying on bag-of-words models and TF-IDF for feature extraction, which lack semantic depth (Li et al., 2022). Additionally, these methods face challenges in capturing underlying information in longer news texts, highlighting limitations in handling complex fake news detection tasks.

2.2. Fake news detection based on deep learning

Nasir, Khan, and Varlamis (2021) propose a hybrid deep learning model combining CNN and RNN to improve fake news detection accuracy. The model uses 1D convolutional layers for local feature extraction, followed by max-pooling for downsampling, and LSTM layers to capture long-term dependencies. This approach effectively manages both spatial (via CNN) and sequential features (via RNN), representing an innovative advancement in fake news detection. Aslam, Ullah Khan, Alotaibi, Aldaej, and Aldubaikil (2021) propose a deep learning ensemble model for fake news detection. The model combines two different deep learning architectures: Bi-LSTM-GRU and Dense neural networks, to handle different types of features. For structured text data, Bi-LSTM-GRU is utilized to extract contextual information comprehensively. For structured data, Dense neural networks are employed. Sastrawan, Bayupati, and Arsa (2022) propose a deep learning model combining CNN, BiLSTM, and residual networks for fake news detection (Li, Li, & Yang, 2021). It incorporates back-translation-based data augmentation (translating English to German and back) to generate additional data, reducing class imbalance and enhancing the model's ability to classify diverse news categories. Palani and Elango (2023) present a content-based deep learning framework, BBC-FND (BERT - BiLSTM - CNN), for detecting fake news on online social networks. The framework combines three models: BERT, BiLSTM, and CNN, to enhance the accuracy and efficiency of detection. News texts are first transformed into word embeddings using BERT, followed by the extraction of global and local information using BiLSTM and CNN, respectively. Mallik and Kumar (2024) introduce a deep learning approach for fake news detection on social media, combining Word2Vec embeddings with LSTM networks to establish a context-independent framework. Pre-trained Word2Vec models generate vector representations, while stacked LSTM layers process these vectors to extract key features of news articles. Ma et al. (2023) introduce the DC-CNN model (Dual-channel Convolutional Neural Networks with Attention-pooling) for social media fake news detection, incorporating DWtext, a dynamic word embedding method to reduce semantic ambiguity in Chinese text. DC-CNN replaces the standard CNN pooling layer with a dual-channel pooling layer, combining Max-pooling and Attention-pooling to better retain local-global feature correlations.

The deep learning methods mentioned above frequently use CNN and RNN models, highlighting their importance in fake news detection.

RNNs capture global information in text, while CNNs focus on local details (Sharifani & Amini, 2023), making their combination effective for extracting comprehensive information from news texts. For word embedding, these methods mainly use Word2vec and BERT, enabling complex semantic extraction beyond traditional machine learning. Despite these advancements, deep learning-based fake news detection still faces challenges:

(1) Existing methods have not taken into account the issue of sample imbalance in fake news datasets, which can adversely affect prediction accuracy.

(2) Existing methods have overlooked the role of residual connections in deep learning networks. Residual connections can mitigate overfitting and information loss problems in deep networks.

(3) Existing methods have not considered the mutual influence between news articles. Taking into account information from other news samples can lead to more accurate judgments.

Based on the above, the paper introduces the AGRC-RCNN model for fake news detection, the details of which will be described in Section 4.

3. Preliminary

3.1. Electra

Electra (Clark, Luong, Le, & Manning, 2020) is a new pre-trained language model built upon the mechanism of self-attention and the Transformer architecture. Unlike other pre-trained language models such as BERT and GPT, Electra adopts an alternative pre-training task called the “Replacing Token Detection Task.” In this task, the model receives a piece of text sequence and replaces some words within it (Zhang, Yu, & Zhu, 2022). The model then needs to identify which words have been replaced. Compared to other pre-training tasks like language model masked filling, Electra’s replacing token detection task is more efficient because the model only needs to learn to recognize the positions of replaced words without predicting all words.

The architecture of the Electra model is similar to BERT but with some key differences. Electra consists of a small masked language model (MLM) (Wettig, Gao, Zhong, & Chen, 2023) called Generator and a Discriminator, which is a binary classifier used to identify real vocabulary and replaced vocabulary in the input sequence. The Generator is responsible for generating the replaced vocabulary, while the Discriminator distinguishes between real vocabulary and generated replacement words. Through this structure, Electra can effectively utilize large-scale corpora for pre-training and has achieved significant performance improvements across various natural language processing tasks. Electra not only excels in model performance but also is more computationally efficient, making it one of the prominent models in the field of natural language processing today.

3.2. LSTM

LSTM is a type of recurrent neural network designed to effectively learn and remember long-term dependencies in sequential data. The structure of LSTM is shown in Fig. 1.

LSTM controls the retention and discard of sequence information through forget gates, input gates, and output gates, thereby mitigating the risks of gradient vanishing and explosion in long sequences (Zhai, Zhang, Fang, & Yao, 2023).

3.3. Self-attention

The self-attention mechanism is a commonly used feature selection method in deep learning (da Costa, Oliveira, & Fileto, 2023). Its main purpose is to allow the model to assign different attention weights to each element in a sequence when processing it, so that the model can

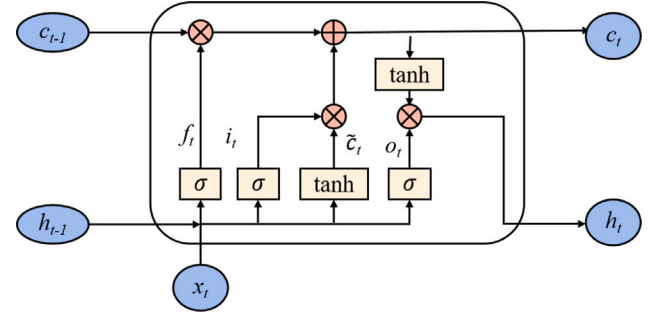


Fig. 1. The structure of LSTM.

focus its attention on the most important parts of the current task. The formula for self-attention is shown in Eq. (1).

$$Attention(Q, K, V) = softmax\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (1)$$

Q, K, and V respectively represent queries, keys, and values, which are inputted as vectors and need to be multiplied by learnable parameter matrices.

3.4. Focal loss

The Focal Loss (Lin, Goyal, Girshick, He, & Dollár, 2017) is a type of loss function used in deep learning to address class imbalance issues, and it is an improvement over the cross-entropy loss function. By introducing two hyperparameters to adjust the relationship between multiple samples and few samples, the formula is shown as Eq. (2).

$$FL(p_t) = -\alpha_t(1 - p_t)^\gamma \log(p_t) \quad (2)$$

In Eq. (2), p_t represents the predicted probability of the positive class by the model, γ is a tuning factor used to decrease attention on easily classified samples and increase attention on hard-to-classify samples. $\log(p_t)$ is the logarithm of the predicted probability, which is a component of the cross-entropy loss. In the binary classification task, if the weight of positive class samples is α_t , then the weight of negative class samples is $1 - \alpha_t$. By adjusting the value of the balancing factor α_t in the focal loss, different degrees of consideration can be given to positive and negative samples during the training process (Zhu, Dai, Hu, & Li, 2020).

4. AGRC-RCNN

4.1. Model overview

The overall structure of the AGRC-RCNN model is shown in Fig. 2, which includes Electra, MSRCNN, AGRC, and Self Attention.

To gain a more intuitive understanding of the false news detection algorithm process of AGRC-RCNN, the algorithm process is presented here as Algorithm 1.

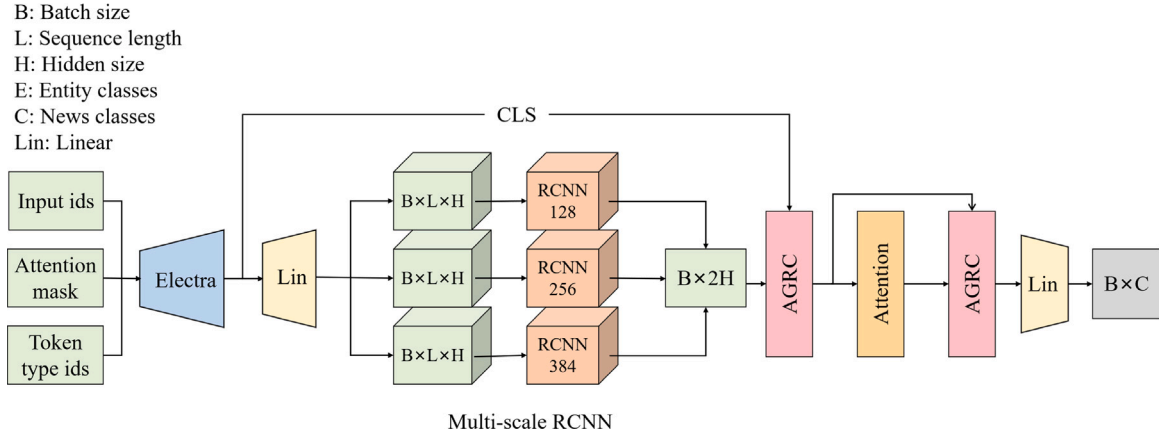


Fig. 2. The structure of AGRC-RCNN.

Algorithm 1 AGRC-RCNN

Input: original dataset(D)
Output: prediction

- 1: Initialization: Electra, RCNN, AGRC, self-attention, Epoch, Mish, Dropout, Linear, Softmax, num_head
- 2: $train, test, eval \leftarrow \text{Divide dataset}(D)$
- 3: **for** $epoch \leftarrow 0$ to $Epoch$ **do**
- 4: $State \leftarrow \text{Electra}(train)$
- 5: $State \leftarrow \text{Linear}(train)$
- 6: **for** $i \leftarrow 0$ to num_head **do**
- 7: $Out_i \leftarrow RCNN_i(State_i)$
- 8: $Out_i \leftarrow \text{Mish}(Out_i)$
- 9: **end for**
- 10: $Out_{msrcnn} \leftarrow \text{Concat}(Out_1, Out_2, \dots, Out_{num_head})$
- 11: $Out_{msrcnn} \leftarrow \text{AGRC}(State[0], Out_{msrcnn})$
- 12: $Out_{att} \leftarrow \text{self-attention}(Out_{msrcnn})$
- 13: $Out_{att} \leftarrow \text{Mish}(Out_{att})$
- 14: $Out_{att} \leftarrow \text{AGRC}(Out_{msrcnn}, Out_{att})$
- 15: $Out \leftarrow \text{Dropout}(Out_{att})$
- 16: prediction $\leftarrow \text{Softmax}(\text{Linear}(Out))$
- 17: **end for**
- 18: **return** prediction

The process of false news detection with AGRC-RCNN is as follows:

(1) Removing stop words and other irrelevant symbols from the dataset. Dividing the dataset into training, testing, and validation sets, and trimming or padding sentences according to the average length of dataset samples. Converting the data into encodings using Electra's Tokenizer tool and packaging the data into batches according to the set batch size.

(2) Using Electra, the news text is converted into word embedding vectors. Then, the news text vectors are inputted into Multi-Scale RCNN to sufficiently extract both global and local information. Combining with the AGRC module for residual connections, it alleviates the issue of gradient vanishing or exploding in the model.

(3) Introduce self-attention mechanism to calculate attention scores between news articles, enabling the sharing of information among them. Then, connect residual information via AGRC, and obtain the prediction results of fake news through Linear and Softmax. Finally, the sample relationships of the balanced dataset are combined with focal loss.

4.2. Multi-scale RCNN

In text classification, RCNN (Lai, Xu, Liu, & Zhao, 2015) is a model based on CNNs and RNNs. It combines the strengths of CNNs in capturing local features and RNNs in modeling sequential information (Lin, Fu, Mao, Wei, & Li, 2019). The structure of RCNN is illustrated in Fig. 3.

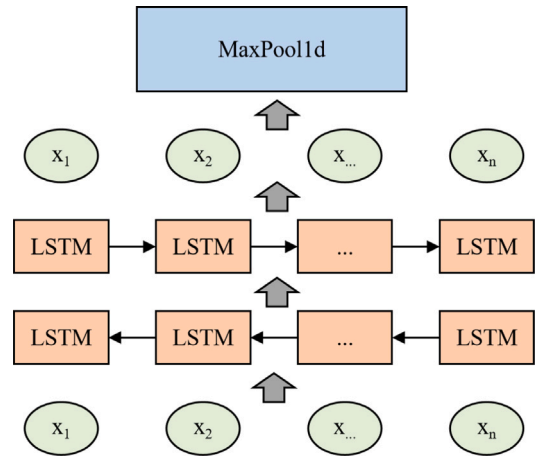


Fig. 3. The structure of RCNN.

As shown in Fig. 3, this paper constructs an RCNN network by combining BiLSTM with maximum pooling. BiLSTM is a variant of RNN that introduces gating mechanisms based on RNN, enabling adaptive adjustment of sequence information propagation. BiLSTM stacks two LSTM layers in opposite directions, allowing consideration of news text features from both forward and backward directions. In the task of detecting fake news, the goal of BiLSTM is to extract deep contextual and global information from news text.

MaxPool1d is a one-dimensional pooling operation used in deep learning to downsample input data, thereby reducing the dimensionality and complexity of the data while preserving important feature information (Zhou, Zhou, Yu, Zhao, & Pan, 2020). MaxPool1d operates by sliding a fixed-size window (or kernel) along the input sequence. The size of the window is a predefined hyperparameter, such as 2, 3, or larger. Thus, based on such windows, MaxPool1d can extract local feature information from news text.

To enable the RCNN to capture context information from different perspectives and scales, this paper constructs a multi-scale RCNN as shown in Fig. 4.

From Fig. 4, it can be seen that the structure of the MSRCNN draws inspiration from the concept of multi-head self-attention networks in Transformers. The text feature sequence first undergoes linear transformations to obtain features for multiple heads. Each head's features are then fed into RCNNs of different scales to extract contextual information, with the outputs of the multi-head networks ultimately being concatenated. The MSRCNN consists of three separate RCNNs, each with hidden layers of sizes 128, 256, and 384, respectively. The final

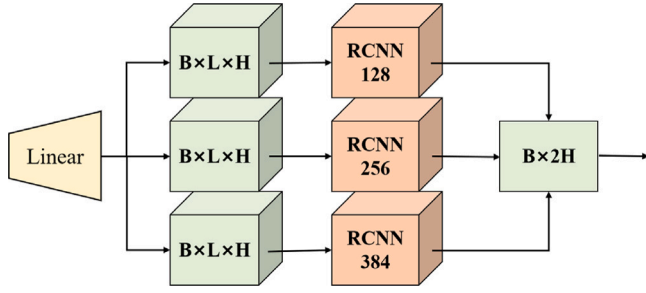


Fig. 4. The structure of MSRCNN.

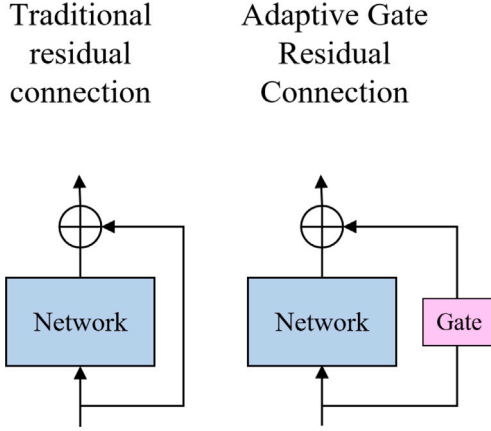


Fig. 5. Traditional residual and Adaptive Gate Residual.

concatenation operation aligns the output vector of the MSRCNN with the overall hidden layer size of the model. This approach allows model to obtain contextual information from news text at different spatial and scale levels, which aids the model in gaining a more comprehensive understanding of the deep-level information within the news text.

4.3. Adaptive Gate Residual Connection

In deep learning, Residual Connection (Cao, Wang, Si, Huang, & Xiao, 2022) is a network architecture technique that allows a layer in the network to directly access the output of a previous layer. The core idea of residual connection is to address the degradation problem in training deep neural networks, where as the network gets deeper, its performance may not improve and could even deteriorate. This is mainly because gradients in deep networks become very small or very large during training, making it difficult for the network to learn (Shin, Cho, Ko, & Gu, 2021). Traditional residual connections directly add previous information to current information, without considering whether the current network needs to receive excessive previous information. If the network does not face the risk of gradient vanishing or exploding, filtering too much previous information can confuse the current feature information, leading to a decline in final performance. To address this issue, this paper introduces a gating mechanism to control the reception of residual information, as shown in Fig. 5.

The gate mechanism in Fig. 5 is implemented through Eq. (3).

$$gate = \text{sigmoid}([residual, x] \times W) \quad (3)$$

In Eq. (3), W is a learnable parameter matrix. The idea of gate control in AGRC comes from LSTM. By concatenating current information with previous information and then multiplying by W , similarity scores are obtained. These similarity scores are passed through a sigmoid function to obtain probability values ranging from 0 to 1. Continuing,

```
class AGRC(nn.Module):
    def __init__(self, in_s, out_s):
        super(AGRC, self).__init__()
        self.w = nn.Linear(in_s, out_s)
        self.sigmoid = nn.Sigmoid()
        self.gate = None

    def forward(self, x, residual):
        pair = torch.cat((x, residual), -1)
        pair = self.w(pair)
        gate = self.sigmoid(pair)
        return x + gate * residual
```

Fig. 6. The PyTorch code of AGRC.

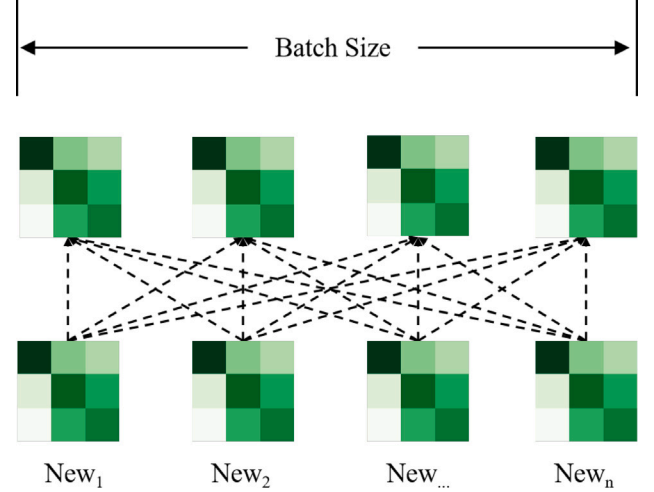


Fig. 7. The computation of self-attention at the level of news documents.

through Eq. (4), the extent to which prior information needs to be considered is controlled.

$$x = gate \times residual + x \quad (4)$$

The Adaptive Gate Residual Connection can be easily implemented with the Pytorch code in Fig. 6.

4.4. Self-attention in news document

In fake news detection, self-attention is often used to calculate the relationship between words, aiming to capture key vocabulary. However, attention between news articles is often overlooked. In the real world, assessing the authenticity of a piece of news typically requires comparison with other news articles. If many related news articles have been verified as true, then the content of this news can be easily identified as a rumor. With this consideration in mind, this paper utilizes self-attention mechanisms to calculate attention scores between news articles, enabling mutual influence between them. The self-attention calculation between news items is illustrated in Fig. 7.

4.5. Focal loss in fake news detection

To alleviate the issue of uneven sample distribution in the fake news detection dataset, this paper employs the focal loss function to adjust the relationship between abundant and scarce samples, enabling the model to achieve optimal balance during the training process.

Table 1
The datasets information.

Dataset	Training	Test	Val	Classes	Avg length
NewsHeadline ^a	21 367	2670	2672	2	10
Covid ^b	6420	2140	2140	2	14
Politifact ^c	17 054	2132	2132	2	483

^a <https://github.com/rishabhmisra/Sarcasm-Detection-using-NN/tree/master>

^b https://huggingface.co/datasets/nany1025/covid_fake_news

^c https://huggingface.co/datasets/Cartinoe5930/Politifact_fake_news

Table 2
The dataset sample distribution table.

Dataset	True	Fake
NewsHeadline	56.3%	43.7%
Covid	52.3%	47.7%
Politifact	57%	43%

5. Experimental results and analysis

5.1. Experimental datasets

The article experiments on three publicly available fake news detection datasets, NewsHeadline (Misra & Arora, 2023), Covid (Patwa et al., 2021), and Politifact. The detailed information of the dataset is shown in Table 1.

From Table 1, it can be seen that the dataset for fake news detection used in this paper includes both long texts and short texts. Therefore, selecting NewsHeadline, Covid, and Politifact can better summarize the field of fake news detection. More detailed information can be obtained from the dataset's website.

As shown in Table 2, the dataset for detecting fake news commonly exhibits the issue of sample imbalance. Positive samples are predominant, while negative samples are in the minority. Introducing focal loss can alleviate this problem, thereby improving the prediction accuracy of fake news detection tasks.

5.2. Experimental evaluation index

Detecting fake news belongs to the category of text classification. Therefore, the paper selects accuracy and F1 score, commonly used in text classification, as the evaluation metrics for the experiments (Umer et al., 2023). The equations are presented as Eqs. (5), (6), (7), and (8).

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (5)$$

$$\text{Precision} = \frac{TP}{TP + FP} \quad (6)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (7)$$

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (8)$$

In the equation above, TP stands for True Positives, TN stands for True Negatives, FP stands for False Positives, and FN stands for False Negatives. In addition, to better evaluate the performance on imbalanced datasets, this paper employs the Macro method to calculate the F1 score.

5.3. Experimental setting

The method selected a learning rate of $5e-6$ for training with a batch size of 64. Default training runs for 30 epochs, and training stops early if there is no decrease in loss on the validation set for 2000 consecutive batches. Sentence lengths in the dataset are cropped according to their average length. Adam is chosen as the optimizer with a dropout rate of 0.1. The gamma value in the focal loss is set to 2.

5.4. Baselines

Machine learning. Naive Bayes (NB), Multilayer Perceptron (MLP), and K-Nearest Neighbors (KNN) are commonly used classifiers in the task of detecting fake news, as they possess good feature selection capabilities. Therefore, this paper selects NB, MLP, and KNN to represent the machine learning approaches for fake news detection. The aforementioned methods uniformly convert news texts into feature vectors using TF-IDF.

Deep learning. In the task of detecting fake news, commonly used deep learning methods include CNN, BiLSTM, GCN (Yao, Mao, & Luo, 2019) (Graph Convolutional Networks), FastText (Joulin et al., 2016), BERT, and RoBERTa (Robustly optimized BERT approach). Meanwhile, they are also the most commonly used deep learning methods in the field of text classification. Among them, CNN and BiLSTM use GloVe¹ word embeddings. BERT² and RoBERTa³ load pre-trained weights, with weight files sourced from Huggingface.

Related works. The paper selects research most relevant to AGRC-RCNN as the comparative methods, which are Bi-LSTM-GRU (Aslam et al., 2021), CNN-RNN (Sastrawan et al., 2022), and Word2vec+LSTM (Mallik & Kumar, 2024). By contrasting them, it is possible to fully highlight the superiority of the proposed method in the field of detecting fake news.

5.5. Comparative experiment

To thoroughly validate the superiority of the proposed method in the task of detecting fake news, comparative experiments were conducted between AGRC-RCNN and baseline methods, as shown in Table 3.

It can be observed from Table 3 that the proposed method has achieved the highest classification accuracy, recall rate, and F1 score across all three datasets (NewsHeadline, Covid, and Politifact). This indicates that the method performs optimally in the task of fake news detection and holds a significant advantage over other methods. This is primarily attributed to the focal loss function utilized by the proposed method, which existing related methods have not combined to address the issue of dataset imbalance. Furthermore, the AGRC network introduced by the proposed method effectively mitigates the risk of model overfitting and enhances the model's feature expression capability. More importantly, the language model used in this paper is Electra, which possesses text feature extraction capabilities superior to those of Word2vec and BERT.

Compared to traditional machine learning methods, deep learning approaches (such as CNN, BiLSTM, GCN, BERT, RoBERTa, etc.) generally perform better in most cases. This indicates that for the task of fake news detection, deep learning methods are capable of capturing features and semantic information more effectively, thereby enhancing classification performance.

It is noteworthy that for small-scale datasets, traditional machine learning methods have achieved superior predictive performance over

¹ <http://nlp.stanford.edu/data/glove.6B.zip>

² <https://huggingface.co/google-bert/bert-base-uncased>

³ <https://huggingface.co/FacebookAI/roberta-base>

Table 3
Classification accuracy on NewsHeadline, Covid, and Politifact (%).

Method	NewsHeadline			Covid			Politifact		
	Accuracy	Recall	F1	Accuracy	Recall	F1	Accuracy	Recall	F1
NB	81.21	79.98	80.34	91.78	90.96	91.33	68.20	65.25	65.14
KNN	76.09	74.26	74.17	89.67	89.39	89.57	65.06	66.68	65.32
MLP	83.55	82.06	83.17	94.49	93.64	94.47	71.05	71.22	71.06
CNN	86.90	86.58	86.73	92.62	92.65	92.61	74.16	74.30	73.90
BiLSTM	86.15	85.88	85.99	91.50	91.44	91.47	68.25	68.57	68.04
GCN	87.05	87.11	87.26	93.64	93.24	93.55	74.03	74.11	74.32
FastText	83.24	83.06	82.88	92.18	92.34	92.11	70.59	70.38	70.06
BERT	89.18	87.66	87.86	93.83	93.62	93.48	75.52	72.03	71.19
RoBERTa	89.33	88.35	88.11	94.06	93.88	93.69	76.88	75.67	76.12
Bi-LSTM-GRU	90.06	89.54	89.44	94.71	94.67	94.63	78.85	78.77	78.60
CNN-RNN	89.94	89.63	89.31	94.71	94.63	94.55	78.34	78.42	78.47
Word2vec+LSTM	89.41	89.38	89.17	94.56	94.47	94.41	78.62	78.54	78.38
Proposed method	91.63	90.76	90.88	95.96	95.17	95.23	79.63	79.23	79.36

Table 4
Sustainability metrics across different models.

Model	NewsHeadline	Covid	Politifact
Proposed method	89.74	90.93	60.61
Bi-LSTM-GRU	88.99	90.23	60.33
CNN-RNN	88.90	89.65	60.82
Word2vec+LSTM	87.69	89.53	60.54

Table 5
Accuracy metrics across different models.

Model	NewsHeadline	Covid	Politifact
Proposed method	96.17	97.10	82.01
Bi-LSTM-GRU	95.47	96.93	81.36
CNN-RNN	96.02	96.85	81.09
Word2vec+LSTM	95.38	93.36	79.94

deep learning methods. For instance, in the Covid dataset, MLP achieved an accuracy of 94.49%, which is better than most deep learning methods. The superior performance of traditional machine learning methods on small-scale datasets may be attributed to the fact that the characteristics of these datasets are more suitable for traditional machine learning algorithms. It is possible that the features of these datasets possess good linear separability, or that traditional machine learning algorithms are more adept at capturing these features. However, when the scale of the dataset increases and the tasks become more complex, traditional machine learning methods may fall short. This is because, in the context of large-scale datasets and complex tasks, the data often exhibits a greater number of features and more intricate relationships, which traditional machine learning methods may not be able to effectively capture. In contrast, deep learning methods, through hierarchical feature learning, are better equipped to handle complex data representations.

As illustrated in Fig. 8, the Receiver Operating Characteristic (ROC) curves across three datasets were generated to further validate the superior performance of AGRC in this domain. The De Long's test (DeLong, DeLong, & Clarke-Pearson, 1988) confirms that AGRC's AUC is statistically higher than that of the comparison models ($p < 0.05$), that providing strong empirical evidence for its superior performance on imbalanced datasets.

To further validate the Sustainability, Accuracy and Explainability of our proposed model, we conducted a comparative analysis of various models utilizing the SAFE metrics (Giudici, 2024; Giudici & Raffinetti, 2023). The experimental results for Sustainability, Accuracy, and Explainability in SAFE are presented in Tables 4, 5 and 6, respectively.

As demonstrated in Tables 4, 5 and 6, while the proposed methodology exhibits superior performance in terms of sustainability and accuracy metrics, it demonstrates relatively inferior results with respect to Explainability measures.

Table 6
Explainability metrics across different models.

Model	NewsHeadline	Covid	Politifact
Proposed method	66.61	67.09	57.35
Bi-LSTM-GRU	66.66	67.11	57.41
CNN-RNN	66.48	67.06	57.53
Word2vec+LSTM	66.57	67.04	56.91

5.6. Ablation experiments

To thoroughly verify the structural effectiveness of each module of the proposed method, ablation experiments will be conducted on AGRC-RCNN. The detailed experimental results and analysis are presented in Appendix.

5.7. Hyperparameter experiments

To determine the optimal value of α in focal loss, the influence of α on prediction accuracy is investigated. Experimental analysis is conducted here with different values of α , and the experimental results are illustrated in Fig. 9.

In Fig. 9, the alpha value refers to the weight of the minority samples. According to the experimental results, there is a very strong correlation between the experimental accuracy and the alpha value. From the trajectory of the discount graph, it is not apparent that the accuracy follows a regular pattern across different alpha values. However, on three datasets, the highest accuracy values correspond to alpha values of 0.9, 0.5, and 0.6, respectively, which correspond to their actual sample distribution situations. When minority samples are assigned higher alpha weights, the accuracy of the fake news detection task begins to increase, but an excessively high alpha value can also lead to a decrease in accuracy.

Ultimately, it was determined that in the NewsHeadline dataset, the focal loss function achieves the best performance with an alpha value of 0.9, reaching an accuracy of 90.53%. In the Covid dataset, the focal loss function performs optimally with an alpha value of 0.5, attaining an accuracy of 94.81%. Within the Politifact dataset, the focal loss function yields the highest accuracy when the alpha value is set at 0.6, amounting to 78.75% accuracy. The influence of the number of heads in MSRCNN on the performance of fake news detection is shown in Fig. 10.

As can be seen from Fig. 10., when the number of heads in MSRCNN is increased to 3, the accuracy of MSRCNN on NewsHeadline is the highest, reaching 91.63%. When the number of heads exceeds 3, the accuracy of the model begins to decline. This suggests that when the number of heads is too few, MSRCNN cannot learn sufficient news information, and when the number of heads is too many, it increases the complexity of the model, leading to a decrease in final predictive performance. Therefore, it is concluded that having 3 heads in MSRCNN is a preferable choice.

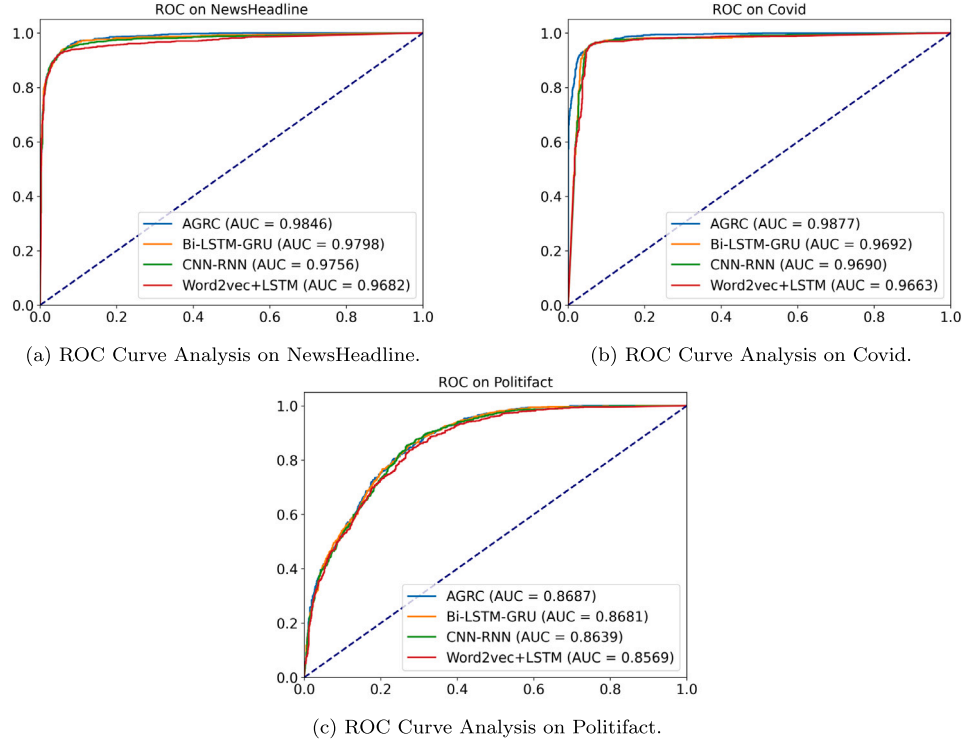
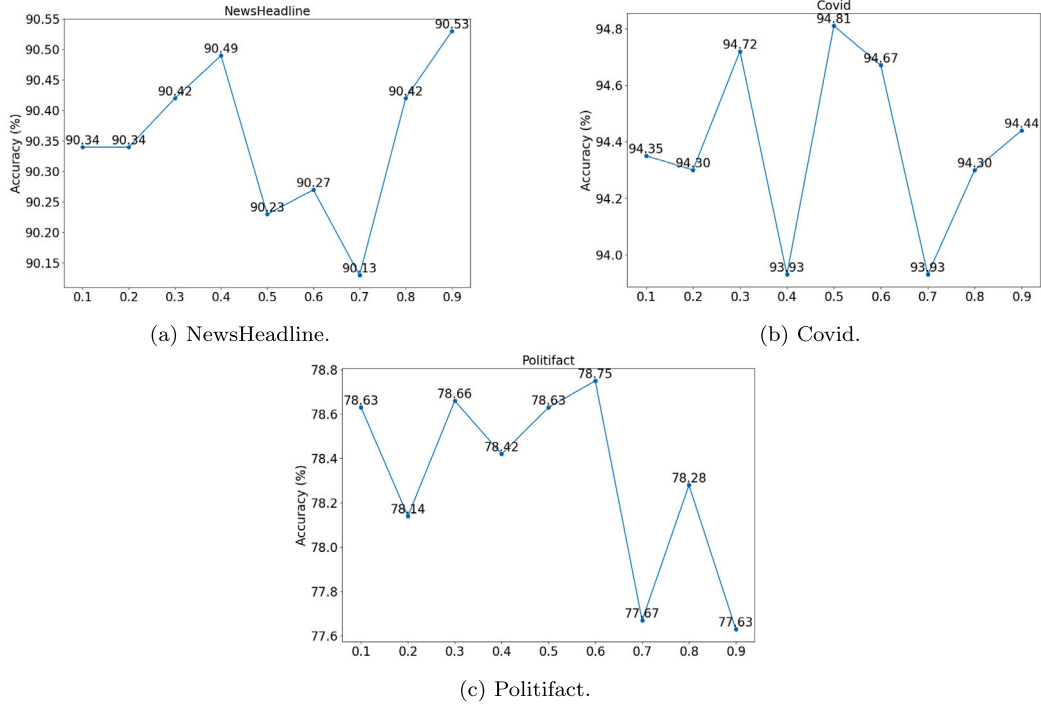


Fig. 8. ROC Curve Analysis.

Fig. 9. The different α value accuracy results.

5.8. Visualization results

To more intuitively demonstrate the superiority of the proposed method in the task of fake news detection, the following will present a visualization result graph. Fig. 11 is the visualization of the gate in AGRC, with data selected from a sample in NewsHeadline.

As shown in Fig. 11, after training, the gate weight matrix will focus on the significant values within the feature information. The gate matrix enables the model to adaptively select which information to retain and which to discard. Fig. 12 is a heatmap of the self-attention scores between five random news samples in NewsHeadline.

As depicted in Fig. 12, proposed model pays more attention to the genuine news samples, using the information from real news to judge

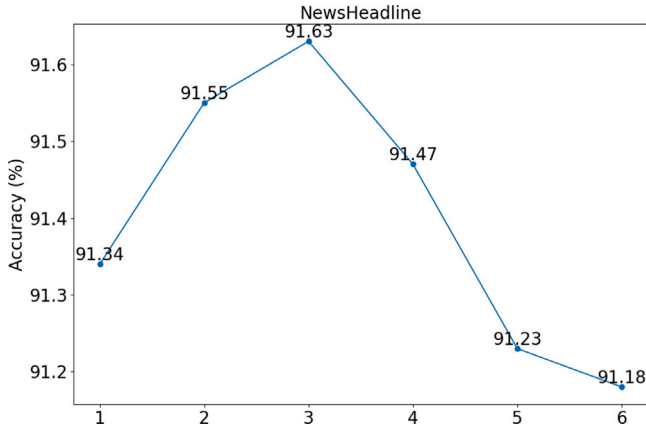


Fig. 10. The relationship between the number of heads in MSRCNN and the accuracy on the NewsHeadline dataset (%).

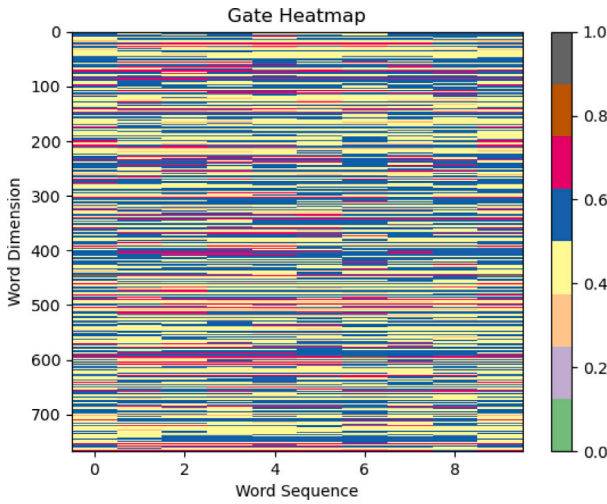


Fig. 11. The visualization graph of the gate matrix after training.

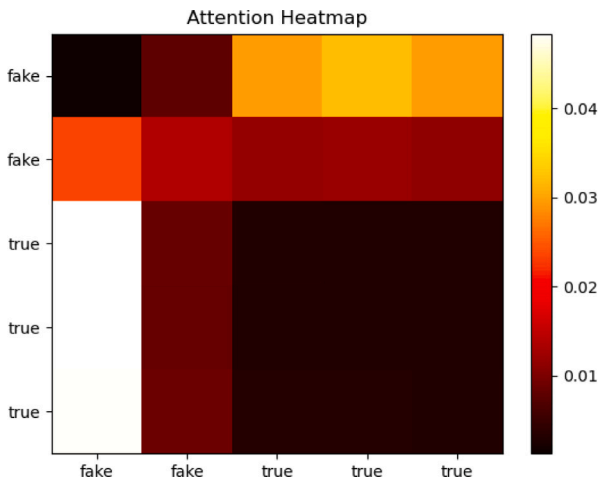


Fig. 12. The self-attention score matrix between news document.

fake news. Additionally, the computation of self-attention between news items allows for the mutual absorption of features from other news, thereby leading to a more comprehensive judgment.

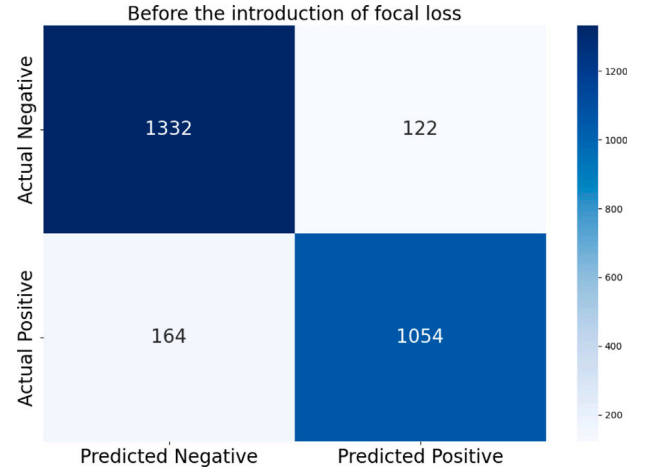


Fig. 13. Before the introduction of focal loss.

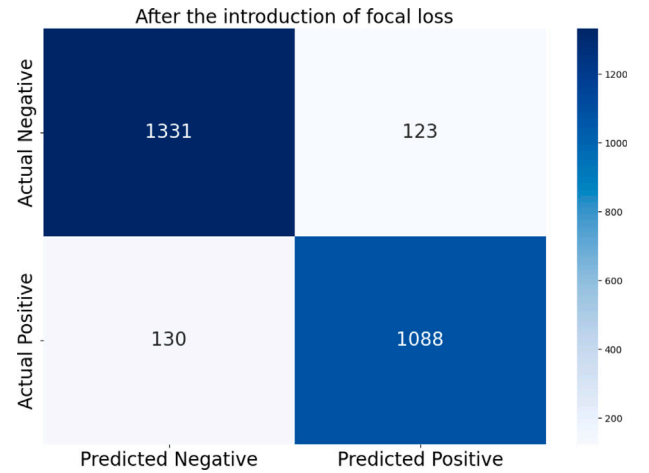


Fig. 14. After the introduction of focal loss.

Figs. 13 and 14 depict confusion matrix visualizations on NewsHeadline. Fig. 13 presents the experimental results on Electra, while Fig. 14 illustrates the results of Electra + Focal Loss.

From the confusion matrices in Figs. 13 and 14, it can be observed that the number of correctly predicted fake news instances belonging to the minority class significantly increased after the introduction of focal loss, rising from 1,054 to 1,088. This clearly demonstrates that focal loss enhances the model's learning capability regarding minority classes, improves the prediction effectiveness for minority classes, and ultimately boosts the overall accuracy.

6. Conclusion

This paper proposes a method for detecting fake news based on adaptive gated residual connections and focal loss, addressing the issues of imbalanced datasets and neglecting the relationships between news. Firstly, the method extracts word embeddings through Electra to enhance the quality of word vectors in news texts. Secondly, RCNN is introduced to further extract both global and local information from news texts, enhancing the model's understanding of news content. An adaptive gating mechanism is employed to control which information the model retains or discards during training. A self-attention mechanism is introduced to calculate the attention between news, allowing news to influence each other and thus make more comprehensive judgments. Finally, the focal loss function is used to enhance the contribution of fewer samples to training. Comparative experiments and

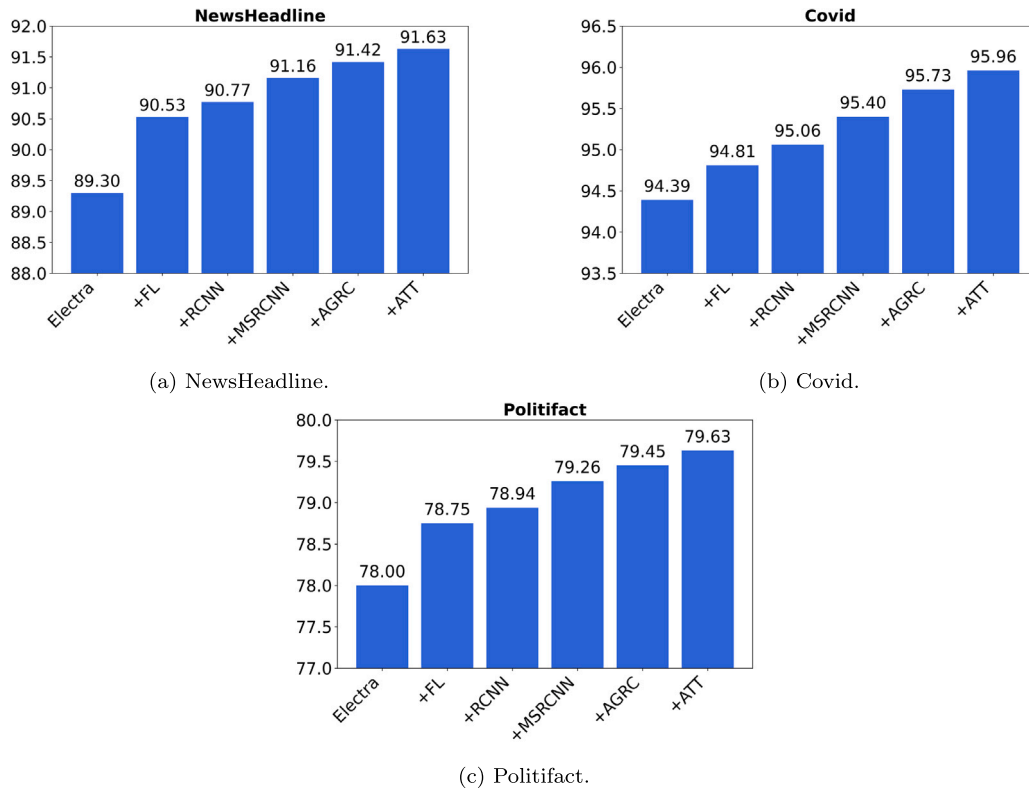


Fig. 15. The accuracy of ablation experiments for each module in the proposed method (%).

ablation experiments are conducted on three public datasets, and the results show that the proposed method achieves the highest prediction accuracy, and all modules of AGRC-RCNN are effective.

The results of the ablation experiments are shown in Fig. 15. The “+” symbol represents the introduction of the current module based on the previous module.

The limitation of this article lies in the absence of employing a more intelligent focal loss method. The optimal hyperparameters for focal loss often require numerous experiments to determine, which is very time-consuming. Future work will focus on designing more advanced focal loss methods to achieve adaptive hyperparameter settings.

CRedit authorship contribution statement

QunHui Zhou: Study conception and design, Data collection, Analysis and interpretation of results, Draft manuscript writing. **Tijian Cai:** Study conception and design, Data collection, Analysis and interpretation of results, Draft manuscript writing.

Generative AI Statement

During the preparation of this work the author(s) used OpenAI ChatGPT 4o in order to improve readability by improving sentence structures, transitions, and grammar. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the publication.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix. Ablation experiments

When comparing Electra to BERT, Electra achieves significantly higher prediction accuracy than BERT. This clearly demonstrates the effectiveness of the proposed approach in utilizing Electra as a language model, enabling Electra to generate higher-quality vectors for news text.

Introducing focal loss on Electra resulted in an increase in the accuracy of fake news detection tasks by 1.23%, 0.42%, and 0.75% on three datasets respectively. This clearly demonstrates the effectiveness of focal loss in addressing the issue of imbalanced dataset samples. Focal loss achieves this by increasing the weight of minority samples in the loss function, thereby enhancing the model’s focus on these minority samples during training, consequently improving the overall prediction accuracy.

After the introduction of RCNN, the accuracy of fake news detection increased by 0.24%, 0.25%, and 0.19% respectively. This indicates that RCNN can further extract contextual information embedded in news texts, enhancing the model’s understanding of news content.

After the introduction of MSRCNN, the accuracy of fake news detection increased by 0.39%, 0.34%, and 0.32% respectively. It has been demonstrated that MSRCNN is capable of understanding the contextual information of news text from different angles and scales, enhancing

the comprehension of the news content, and thus making more accurate news predictions.

After introducing AGRC, the accuracy of fake news detection increased by 0.26%, 0.33%, and 0.19% respectively. This illustrates that gated residual connections can adaptively adjust the extent to which residual information needs to be considered, thus avoiding potential information forgetting in the model. It also prevents the impact on model performance caused by considering too much residual information by adaptively adjusting the required residual information through learnable parameter matrices.

After introducing the self-attention mechanism, the accuracy increased by 0.21%, 0.23%, and 0.18%, respectively. This explains the calculation of attention scores between news, allowing the model to fully consider the interrelation among news. Before making judgments on the veracity of current news, it fully takes into account the information from other news.

In summary, the experimental results indicate that the proposed method is effective in its architecture, with each module contributing to the improvement of performance in the task of detecting fake news.

Data availability

Data will be made available on request.

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